Matrix Manifold Precoder Design for Massive MIMO High-Speed Railway Communications with Channel Prediction

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***Abstract*—In high-speed railway (HSR) communications, the channel suffers from severe channel aging effect caused by the high mobility. To address this issue, we investigate the precoder design against channel aging in massive multiple-input multiple-output (MIMO) systems with channel prediction. First of all, we introduce the concept of the quadruple beams (QBs), and establish a QB based channel model with sampled quadruple steering vectors. Then, the upcoming space domain channel can achieve a higher accuracy by channel prediction. We consider the precoder design on the Riemannian submanifold formed by the precoders satisfying the total power constraint (TPC). The Riemannian conjugate gradient (RCG) method is proposed to solve the problem on the manifold. The RCG method mainly involves the matrix multiplication and avoids the need of matrix inversion of the transmit antenna dimension. The simulation results demonstrate the effectiveness of the proposed channel model and the superiority of the RCG method for precoder design against channel aging.**

***Index Terms*—High-speed railway, massive MIMO, manifold optimization, precoding, Riemannian submanifold.**

I. Introduction

With the rapid deployment of the high-speed railway (HSR) throughout the world, communication systems tailored for HSR have gained great attention from both academia and industry [1]. Meanwhile, fueled by the evolution of wireless technologies, the pursuit for increased data rates and enhanced quality services also manifests an exponential growth trend in HSR communications [2]. Massive multi-input multi-output (MIMO) is one of the most important technologies to deliver these urgent demands for the HSR communications [3], [4] by serving numerous users simultaneously on the same time-frequency resource [5].

However, accommodating a number of users simultaneously inevitably causes severe inter-user interference [6]. To deal with this challenge, precoding is used to suppress the interference, and has been widely investigated in the HSR communications [7]. In practice, the downlink precoder design highly depends on the available channel state information (CSI) at the BS side. The channel estimated from the pilot segment becomes outdated quickly due to the severe Doppler effect in HSR communications [8], [9] . To address this issue, We introduce the concept of quadruple beams (QBs) in the space-frequency-time (SFT) domain and build a QB based channel model with sampled quadruple steering vectors [10] within the orthogonal frequency division multiplexing (OFDM) transmission framework . Then, the space domain channel can be predicted via the relationship between the space domain channel and the QB domain channel.

Matrix manifold optimization has been widely investigated thanks to its ability of transforming a constrained problem in Euclidean space into an unconstrained one on a manifold [11], [12], [13]. More importantly, the RCG method can avoid the matrix inversion over the transmit antenna dimension, which is of increasingly benefits in the future wireless networks where more antennas will be equipped at the BS side [14]. Therefore, we are motivated to investigate the RCG method for precoder design for HSR communications.

In this paper, we investigate the matrix manifold precoder design against channel aging for massive MIMO HSR communications with channel prediction. We establish a QB based channel model from the physical principle of HSR channels with sampled quadruple steering vectors in the SFT domain, each of which corresponds to a physical QB consisting of the angle of arrival (AoA) beam, the angle of departure (AoD) beam, the frequency-beam, and the temporal-beam, pointing towards the sampled directional cosine of AoA, directional cosine of AoD, delay, and Doppler frequency, respectively. Then, the forthcoming space domain channel can be predicted against channel aging. The complexity analyses show that our method has a much lower complexity than the existing methods. The simulation results demonstrate the effectiveness of our proposed channel model and validate the superiority of the RCG method for precoder design against channel aging.

II. Channel Modeling and Channel Prediction

A. System Configuration

We consider a single-cell massive MIMO-OFDM system in the HSR scenario, which is dominated by the line of sight (LoS)

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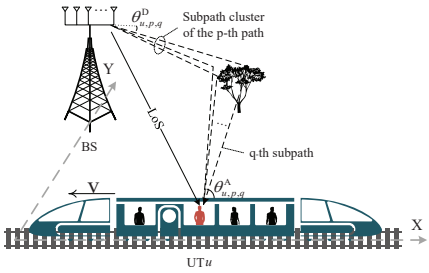


Fig. 1. The massive MIMO system in HSR.

component [15]. The BS serves U UTs in the HSR that moves at the speed of v km/h in the viaduct scenario. $\mathcal{U} = \{1, 2, \cdots, U\}$ represents the set of all the UTs. There are M_t and M_r uniform linear antennas (ULA) equipped at the BS side and each UT side, respectively. As shown in Fig. 1, there are totally P_u paths from the BS to UT u, including one LoS path. The p-th path is composed of $Q_{p,u}$ subpaths, which are termed as the subpath cluster of path p.

Each subframe consists of N_b time slots and each slot is made up of N_s OFDM symbols. Let $N = N_b N_s$ denote the total number of the OFDM symbols in a subframe. Each OFDM symbol has N_c subcarriers, where N_v valid sub-carriers are used for data transmission with the index set of $\mathcal{S}_v = (m_0, m_1, \cdots, m_{N_v-1})$. Let f_c, $\lambda_c = \frac{c}{f_c}$ and Δf represent the center frequency, wave length and the subcarrier spacing, respectively. The system sampling interval is $T_s = \frac{1}{N_c \Delta f}$ and the length of cyclic prefix (CP) is N_g. $T_{\mathrm{sym}} = (N_c + N_g) T_s$ denotes the time duration of an OFDM symbol including CP.

Assuming the channel parameters, including the Doppler shift, time delay, AoD and AoA, are time invariant in each subframe. In addition, we assume that the CSI keeps constant within an OFDM symbol time duration and varies symbol by symbol due to the severe Doppler effect. In each slot, the uplink sounding reference signals are sent once at the n_{p}-th OFDM symbol, and the rest symbols are used for downlink (DL) transmission. Assuming that the CSI is perfectly known at the user side via the DL training phase.

B. QB Based Channel Model

Let $\mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right)$ denote the user antenna response of the m_r-th antenna of UT u along the q-th subpath of the p-th path corresponding to the AoA $\theta^A_{u,p,q}$, where $\Omega^A_{u,p,q} = \cos\left(\theta^A_{u,p,q}\right)$. Let $\mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right)$ signify the BS antenna response of the m_t-th antenna of UT u along the q-th subpath of p-th path corresponding to the AoD with $\Omega^D_{u,p,q} = \cos\left(\theta^D_{u,p,q}\right)$. $\mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right)$ and $\mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right)$ can be written as $\mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right) = e^{-\jmath 2\pi m_r \Delta_r \Omega^A_{u,p,q}}$, $\mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right) = e^{-\jmath 2\pi m_t \Delta_t \Omega^D_{u,p,q}}$, respectively, where Δ_r and Δ_t are the normalized receive antenna separation and transmit antenna separation, respectively. Let $\alpha_{u,p,q}$, $\tau_{u,p,q}$ and $\nu_{u,p,q}$ represent the propagation loss, propagation delay and the Doppler shift of the u-th UT along the q-th subpath of the p-th path, respectively. We assume that all the subpaths of a dominant path share the same time delay and Doppler shift.

Note that the Doppler shift of the LoS path can be easily compensated to mitigate the inter-carrier interference (ICI) [16]. Define $\gamma_{u,p,q} = \alpha_{u,p,q} * e^{-\jmath 2\pi f_c \tau_{u,p}}$ as the complex-valued gain. The channel impulse response between the m_t-th transmit antenna and the m_r-th receive antenna of UT u after Doppler compensation is then given by

$$\bar{h}_{u,m_r,m_t}(t,\tau) = \sum_{p=1}^{P_u}\sum_{q=1}^{Q_{p,u}} \gamma_{u,p,q}\mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right)\mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right) \times e^{\jmath 2\pi \nu^{\mathrm{cps}}_{u,p} t}\delta\left(\tau - \tau_{u,p}\right), \tag{1}$$

where $\nu^{\mathrm{cps}}_{u,p} = \nu_{u,p} - \nu^{\mathrm{LoS}}_u$ is the Doppler frequency after compensation with ν^{LoS}_u being the Doppler shift of the LoS path of UT u. After the OFDM operation, the channel frequency response can be written as

$$h^{\mathrm{SFT}}_{u,m_r,m_t,m,n} = \sum_{p=1}^{P_u}\sum_{q=1}^{Q_{p,u}} \gamma_{u,p,q}\mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right)\mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right) \times e^{-\jmath 2\pi m \Delta f \tau_{u,p}} e^{\jmath 2\pi \nu^{\mathrm{cps}}_{u,p} n T_{\mathrm{sym}}}. \tag{2}$$

Based on (2), we further consider the whole space-frequency-time (SFT) domain channel over N OFDM symbols between the BS and UT u, which is termed as the SFT domain channel vector $\mathbf{h}^{\mathrm{SFT}}_u$, with element $h^{\mathrm{SFT}}_{u,m_r,m_t,m,n}$ of index $(nN_vM_tM_r + (m-m_0)M_tM_r + m_tM_r + m_r)$. Let

$$\begin{aligned}
\mathbf{u}\left(\Omega^A\right) &= [1, e^{-\jmath 2\pi \Delta_r \Omega^A}, \cdots, e^{-\jmath 2\pi (M_r-1)\Delta_r \Omega^A}],\\
\mathbf{v}\left(\Omega^D\right) &= [1, e^{-\jmath 2\pi \Delta_t \Omega^D}, \cdots, e^{-\jmath 2\pi (M_t-1)\Delta_t \Omega^D}],\\
\boldsymbol{f}(\tau) &= \left[e^{-\jmath 2\pi m_0 \Delta f \tau}, \cdots, e^{-\jmath 2\pi m_{N_v-1}\Delta f \tau}\right],\\
\mathbf{d}(\nu) &= \left[1, e^{\jmath 2\pi \nu T_{\mathrm{sym}}}, \cdots, e^{\jmath 2\pi \nu (N-1) T_{\mathrm{sym}}}\right]
\end{aligned} \tag{3}$$

denote the steering vectors in the space domain of the receive antenna, the space domain of the transmit antenna, the frequency domain and the time domain, respectively. Denote

$$\mathbf{s}\left(\Omega^A, \Omega^D, \tau, \nu\right) = \mathbf{d}(\nu) \otimes \left\{\boldsymbol{f}(\tau) \otimes \left[\mathbf{v}\left(\Omega^D\right) \otimes \mathbf{u}\left(\Omega^A\right)\right]\right\} \tag{4}$$

as the SFT domain steering vector. In this way, the SFT domain channel can be expressed as

$$\mathbf{h}^{\mathrm{SFT}}_u = \sum_{p=1}^{P_u}\sum_{q=1}^{Q_{p,u}} \gamma_{u,p,q}\mathbf{s}\left(\Omega^A_{u,p,q}, \Omega^D_{u,p,q}, \tau_{u,p}, \nu^{\mathrm{cps}}_{u,p}\right). \tag{5}$$

The channel parameters $\Omega^A_{u,p,q}$, $\Omega^D_{u,p,q}$, $\tau_{u,p}$ and $\nu^{\mathrm{cps}}_{u,p}$ are limited in the sets $\mathcal{S}_{\mathrm{r}} = \left\{\Omega^A \mid \Omega^A \in [-1, 1)\right\}$, $\mathcal{S}_{\mathrm{t}} = \left\{\Omega^D \mid \Omega^D \in [-1, 1)\right\}$, $\mathcal{S}_{\mathrm{de}} = \left\{\tau \mid \tau \in [0, \tau_{\max})\right\}$ and $\mathcal{S}^u_{\mathrm{do}} = \left\{\nu \mid \nu \in \left[-\nu_{\max} - \nu^{\mathrm{LoS}}_u, \nu_{\max} - \nu^{\mathrm{LoS}}_u\right)\right\}$, respectively, where $\tau_{\max} = \frac{N_g}{N_c \Delta f}$ is the maximum delay spread and $\nu_{\max} = \frac{\mathrm{v}}{\lambda_c} = \frac{N_d}{2NT_{\mathrm{sym}}}$ is the maximum Doppler shift [10]. We uniformly divide these sets into multiple disjoint subsets as $\mathcal{S}_r = \bigcup_{n_r=0}^{N_r-1} \mathcal{S}^r_{n_r}$, $\mathcal{S}^r_{n_r} = [\frac{2n_r - N_r}{N_r}, \frac{2(n_r+1)-N_r}{N_r})$, $\mathcal{S}_t = \bigcup_{n_t=0}^{N_x-1} \mathcal{S}^t_{n_t}$, $\mathcal{S}^t_{n_t} = [\frac{2n_t - N_t}{N_t}, \frac{2(n_t+1)-N_t}{N_t})$, $\mathcal{S}_{\mathrm{de}} = \bigcup_{n_{\mathrm{de}}=0}^{N_{\mathrm{de}}-1} \mathcal{S}^{\mathrm{de}}_{n_{\mathrm{de}}}$, $\mathcal{S}^{\mathrm{de}}_{n_{\mathrm{de}}} = [\frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}} N_v \Delta f}, \frac{N_\tau (n_{\mathrm{de}}+1)}{N_{\mathrm{de}} N_v \Delta f})$, $\mathcal{S}^u_{\mathrm{do}} = \bigcup_{n_{\mathrm{do}}=0}^{N_{\mathrm{do}}-1} \mathcal{S}^{u,\mathrm{do}}_{n_{\mathrm{do}}}$, $\mathcal{S}^{u,\mathrm{do}}_{n_{\mathrm{do}}} = [\frac{N_d\left(n_{\mathrm{do}} - \frac{N_{\mathrm{do}}}{2}\right)}{N_{\mathrm{do}} N T_{\mathrm{sym}}} - \nu^{\mathrm{LoS}}_u, \frac{N_d\left(n_{\mathrm{do}}+1-\frac{N_{\mathrm{do}}}{2}\right)}{N_{\mathrm{do}} N T_{\mathrm{sym}}} - \nu^{\mathrm{LoS}}_u)$, where $N_\tau \triangleq \lceil \frac{N_g N_v}{N_c} \rceil$. $N_r = F_r M_r$, $N_t = F_t M_t$, $N_{\mathrm{de}} = F_{\mathrm{de}} N_\tau$ and $N_{\mathrm{do}} = F_{\mathrm{do}} N_d$, where F_r, F_t, F_{de} and F_{do} are the fine factors. Then, (5) can be approximated

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as

$$\mathbf{h}_u^{\mathrm{SFT}} = \sum_{n_r=0}^{N_r-1}\sum_{n_t=0}^{N_t-1}\sum_{n_{\mathrm{de}=0}}^{N_{\mathrm{de}-1}}\sum_{n_{\mathrm{do}}=0}^{N_{\mathrm{do}}-1}\sum_{\substack{\Omega^A_{u,p,q}\in\mathcal{S}^r_{n_r},\Omega^D_{u,p,q}\in\mathcal{S}^t_{n_t},\\ \tau_{u,p}\in\mathcal{S}^{\mathrm{de}}_{n_{\mathrm{de}}},\nu^{\mathrm{cps}}_{u,p}\in\mathcal{S}^{u,\mathrm{do}}_{n_{\mathrm{do}}}}} \gamma_{u,p,q}\mathbf{s}\left(\Omega^A_{u,p,q},\Omega^D_{u,p,q},\tau_{u,p},\nu^{\mathrm{cps}}_{u,p}\right), \tag{6}$$

where the steering vector $\mathbf{s}\left(\Omega^A_{u,p,q},\Omega^D_{u,p,q},\tau_{u,p},\nu^{\mathrm{cps}}_{u,p}\right)$ is approximated as the sampled steering vector $\mathbf{s}\left(\Omega^A_{n_r},\Omega^D_{n_t},\tau_{n_{\mathrm{de}}},\nu_{n_{\mathrm{do}}}\right)$ with $\Omega^A_{n_r}=\frac{2n_r-N_r}{N_r}$, $\Omega^D_{n_t}=\frac{2n_t-N_t}{N_t}$, $\tau_{n_{\mathrm{de}}}=\frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}}N_v\Delta f}$ and $\nu_{n_{\mathrm{do}}}=\frac{N_d(n_{\mathrm{do}}-N_{\mathrm{do}}/2)}{N_{\mathrm{do}}NT_{\mathrm{sym}}}-\nu_u^{\mathrm{LoS}}$. As the quantities N_r, N_t, N_{de} and N_{do} of partitioned subsets increase, the accuracy of this approximation improves. We then define a transform matrix $\mathbf{S}_u$ that consists of all sampled steering vectors of UT u. $\mathbf{S}_u$ takes the form of $\mathbf{S}_u=\mathbf{D}_u\otimes[\mathbf{F}\otimes(\mathbf{V}\otimes\mathbf{U})]$, where $\mathbf{D}_u\in\mathbb{C}^{N\times N_{\mathrm{do}}}$ with $[\mathbf{D}_u]_{ij}=e^{\jmath 2\pi i\nu_j T_{\mathrm{sym}}}$, $\mathbf{F}\in\mathbb{C}^{N_v\times N_{\mathrm{de}}}$ with $[\mathbf{F}]_{ij}=e^{-\jmath 2\pi(m_0+i)\Delta f\tau_j}$, $\mathbf{V}\in\mathbb{C}^{M_t\times N_t}$ with $[\mathbf{V}]_{ij}=e^{-\jmath 2\pi i\Delta_t\Omega_j^D}$ and $\mathbf{U}\in\mathbb{C}^{M_r\times N_r}$ with $[\mathbf{U}]_{ij}=e^{-\jmath 2\pi i\Delta_r\Omega_j^A}$. In consequence, the SFT domain channel (6) can be expressed by as

$$\mathbf{h}_u^{\mathrm{SFT}}=\mathbf{S}_u\mathbf{h}_u^{\mathrm{QB}}, \tag{7}$$

where $\mathbf{h}_u^{\mathrm{QB}}\in\mathbb{C}^{N_{\mathrm{do}}N_{\mathrm{de}}N_tN_r\times 1}$ is the QB domain channel and $\mathbf{S}_u$ is the transform matrix from the QB domain channel to the SFT domain channel of UT u. To be specific, $\mathbf{h}_u^{\mathrm{QB}}\in\mathbb{C}^{N_{\mathrm{do}}N_{\mathrm{de}}N_tN_r\times 1}$ is given by

$$\left[\mathbf{h}_u^{\mathrm{QB}}\right]_{n_{\mathrm{do}}N_{\mathrm{de}}N_tN_r+n_{\mathrm{de}}N_tN_t+n_tN_r+n_r}=\sum_{\substack{\Omega^A_{u,p,q}\in\mathcal{S}^r_{n_r},\Omega^D_{u,p,q}\in\mathcal{S}^t_{n_t},\\ \tau_{u,p}\in\mathcal{S}^{\mathrm{de}}_{n_{\mathrm{de}}},\nu^{\mathrm{cps}}_{u,p}\in\mathcal{S}^{u,\mathrm{do}}_{n_{\mathrm{do}}}}}\gamma_{u,p,q}. \tag{8}$$

Each column of $\mathbf{S}_u$ corresponds to a physical QB in the SFT domain. Thus, we refer to (7) as the QB based channel model.

Recall that the viaduct scenario is dominated by the LoS path, indicating that the power of $\mathbf{h}_u^{\mathrm{QB}}$ is dominated by the LoS elements, and the viaduct scenario exhibits limited propagation paths and angle spread due to the sparse scattering environment. All these properties lead to $\mathbf{h}_u^{\mathrm{QB}}$ being a sparse vector. Assuming the elements in $\mathbf{h}_u^{\mathrm{QB}}$ follow independent complex Gaussian distributions with zero mean and different variances.

C. Channel Prediction

Let $\hat{\mathbf{h}}_u^{\mathrm{QB}}$ stand for the QB domain channel estimated from the pilot segments, then the corresponding estimated SFT domain channel is given by $\hat{\mathbf{h}}_u^{\mathrm{SFT}}=\mathbf{S}_u\hat{\mathbf{h}}_u^{\mathrm{QB}}\in\mathbb{C}^{NN_vM_tM_r\times 1}$. In a quasi-static scenario, it is typical to directly apply the channel estimated from the pilot segments to the rest of symbols for precoding as the channel nearly keeps constant during a time slot. Let $\mathbf{S}_{u,\mathrm{p}}=\mathbf{D}_{u,\mathrm{p}}\otimes\mathbf{F}\otimes\mathbf{V}\otimes\mathbf{U}$ denote the transform matrix from the QB domain channel to the SFT domain channel at the pilot segments, where $\mathbf{D}_{u,\mathrm{p}}=\mathbf{\Theta}_{\mathrm{p}}\mathbf{D}_u$ with the n_b-th row of $\mathbf{\Theta}\in\mathbb{C}^{N_b\times N}$ being the $(n_bN_s+n_{\mathrm{p}})$-th row of $\mathbf{I}_N$ for $n_b=1,2,\cdots,N_b$. Then, the estimated SFT domain at the pilot segments can be obtained by setting the selection matrix $\mathbf{S}_u$ to $\mathbf{S}_{u,\mathrm{p}}$ and is given by

$$\hat{\mathbf{h}}_{u,\mathrm{p}}^{\mathrm{SFT}}=\mathbf{S}_{u,\mathrm{p}}\hat{\mathbf{h}}_u^{\mathrm{QB}}\in\mathbb{C}^{N_bN_vM_tM_r\times 1}, \tag{9}$$

which can be directly employed for precoder design in the quasi-static scenario. However, the channel changes quickly in HSR communications, necessitating the pursuit for a heightened precise CSI at each symbol without increasing the pilot overhead. In this case, we can take advantage of the relationship between $\mathbf{h}_u^{\mathrm{SFT}}$ and $\mathbf{h}_u^{\mathrm{QB}}$ to predict the channel between pilot segments of interest. Let $\mathbf{S}_{u,n,m}=\tilde{\mathbf{D}}_{u,n}\otimes\tilde{\mathbf{F}}_m\otimes\mathbf{V}\otimes\mathbf{U}$ represent the transform matrix from the QB domain channel to the space domain channel at the m-th subcarrier of the n-th OFDM symbol, where $\tilde{\mathbf{D}}_{u,n}=\mathbf{\Theta}_n\mathbf{D}_u\in\mathbb{C}^{1\times N_{\mathrm{do}}}$ with the n-th element of $\mathbf{\Theta}_n\in\mathbb{C}^{1\times N}$ being 1, and $\tilde{\mathbf{F}}_m=\mathbf{\Phi}_m\mathbf{F}\in\mathbb{C}^{1\times N_{\mathrm{de}}}$ with the m-th element of $\mathbf{\Phi}_m\in\mathbb{C}^{1\times N_v}$ being 1, respectively. By setting the selection matrix $\mathbf{S}_u$ to $\mathbf{S}_{u,n,m}$, the predicted channel at the m-th subcarrier of the n-th OFDM symbol can be written as

$$\hat{\mathbf{h}}_{u,m,n}^{\mathrm{S}}=\mathbf{S}_{u,n,m}\hat{\mathbf{h}}_u^{\mathrm{QB}}[n_b]\in\mathbb{C}^{M_tM_r\times 1}. \tag{10}$$

By reshaping $\hat{\mathbf{h}}_{u,m,n}^{\mathrm{S}}$, we can obtain the channel matrix $\hat{\mathbf{H}}_{u,m,n}\in\mathbb{C}^{M_r\times M_t}$ for precoder design at the m-th subcarrier of the n-th OFDM symbol given by

$$\hat{\mathbf{H}}_{u,m,n}=\mathrm{unvec}\left(\mathbf{S}_{u,n,m}\hat{\mathbf{h}}_u^{\mathrm{QB}}[n_b],(M_r,M_t)\right). \tag{11}$$

III. Problem Formulation and Riemannian Ingredients in Matrix Manifold Optimization

A. Problem Formulation on the Riemannian Submanifold

We investigate the precoder design against channel aging at a given subcarrier, whose subscript is thereby omitted for brevity. Let $\mathbf{x}_{kn}\in\mathbb{C}^{d_k}$ denote the signal transmitted to the k-th UT at the given subcarrier of the n-th symbol satisfying $\mathbb{E}\left\{\mathbf{x}_{kn}\mathbf{x}_{kn}^H\right\}=\mathbf{I}_{d_i}$. The received signal of the k-th UT at the n-th OFDM symbol is given by

$$\mathbf{y}_{kn}=\hat{\mathbf{H}}_{kn}\mathbf{P}_{kn}\mathbf{x}_{kn}+\hat{\mathbf{H}}_{kn}\sum_{\ell\neq i,\ell=1}^{U}\mathbf{P}_{\ell n}\mathbf{x}_{\ell n}+\mathbf{z}_{kn}, \tag{12}$$

where $\mathbf{P}_{kn}\in\mathbb{C}^{M_t\times d_k}$ is the precoding matrix of user k at the n-th OFDM symbol and $\mathbf{z}_{kn}$ is the Gaussian noise distributed as $\mathcal{CN}\left(0,\sigma_z^2\mathbf{I}_{M_r}\right)$. The interference plus noise is treated as the Gaussian noise for the worst case design with the covariance matrix given by $\mathbf{R}_{kn}=\hat{\mathbf{H}}_{kn}\sum_{\ell\neq k}\mathbf{P}_{\ell n}\mathbf{P}_{\ell n}^H\hat{\mathbf{H}}_{kn}^H+\sigma_z^2\mathbf{I}_{M_r}$. Assuming that $\mathbf{R}_{kn}$ is known at the user side, the rate of user k can be written as $\mathcal{R}_{kn}=\mathrm{logdet}(\hat{\mathbf{H}}_{kn}\mathbf{P}_{kn}\mathbf{P}_{kn}^H\hat{\mathbf{H}}_{kn}^H+\mathbf{R}_{kn})-\mathrm{logdet}(\mathbf{R}_{kn})$, and the ergodic weighted sum-rate maximization problem can be formulated as

$$\mathcal{P}_1:\arg\max_{\mathbf{P}_n}\sum_{k\in\mathcal{U}}w_k\mathcal{R}_{kn}\ \ \mathrm{s.t.}\sum_{k\in\mathcal{U}}\mathrm{tr}\left(\mathbf{P}_{kn}^H\mathbf{P}_{kn}\right)=P_t, \tag{13}$$

where P_t is the total power constraint. To avoid the inverses of large dimensional matrices, it is proposed to solve the problem on the manifold [13]. Let $\mathbf{P}_n=[\mathbf{P}_{1n},\cdots,\mathbf{P}_{Un}]\in\mathbb{C}^{M_t\times\sum_{k\in\mathcal{U}}d_k}$ denote the stacked precoding matrix. It has been proved in [13, Theorem 1] that the set

$$\mathcal{M}=\left\{\mathbf{P}_n\mid F(\mathbf{P}_n)=\mathrm{tr}\left(\mathbf{P}^H\mathbf{P}\right)-P_t=0\right\} \tag{14}$$

forms a Riemannian submanifold of the linear manifold $\mathcal{N}:=\mathbb{C}^{M_t\times\sum_{k\in\mathcal{U}}d_k}$ with the Riemannian metric $g\left(\boldsymbol{\xi}_{\mathbf{P}_n},\boldsymbol{\zeta}_{\mathbf{P}_n}\right)=\Re\left\{\mathrm{tr}\left(\boldsymbol{\xi}_{\mathbf{P}_n}^H\boldsymbol{\zeta}_{\mathbf{P}_n}\right)\right\}$, where $\boldsymbol{\xi}_{\mathbf{P}_n}$ and $\boldsymbol{\zeta}_{\mathbf{P}_n}$ are two tangent vectors to $\mathcal{M}$ at $\mathbf{P}_n$. Therefore, the constrained problem (13) can also

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be reformulated as an unconstrained one on the Riemannian submanifold $\mathcal{M}$ as

$$\mathcal{P}_2 : \arg\min_{\mathbf{P}_n \in \mathcal{M}} f(\mathbf{P}_n), \tag{15}$$

where $f(\mathbf{P}_n) = -\sum_{k\in\mathcal{U}} w_k \mathcal{R}_{kn}^{\mathrm{ub}}$ is the objective function.

B. Riemannian Ingredients for Robust Precoder Design

Every point $\mathbf{P}_n$ on the nonlinear $\mathcal{M}$ is attached to a linear tangent space denoted by $T_{\mathbf{P}_n}\mathcal{M}$, which is a set formed by all the tangent vectors to $\mathcal{M}$ at $\mathbf{P}_n$. By introducing the retraction, which is a mapping from the tangent space to the manifold, we can generalize the notion of moving along the tangent vector while remaining on the nonlinear manifold $\mathcal{M}$. The retraction of $\mathcal{M}$ is given by

$$R_{\mathbf{P}_n}^{\mathcal{M}}\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right) = \rho\left(\mathbf{P}_n + \boldsymbol{\xi}_{\mathbf{P}_n}\right) \tag{16}$$

with $\rho = \frac{\sqrt{P_t}}{\sqrt{\mathrm{tr}\left(\left(\mathbf{P}_n+\boldsymbol{\xi}_{\mathbf{P}_n}\right)^H\left(\mathbf{P}_n+\boldsymbol{\xi}_{\mathbf{P}_n}\right)\right)}}$. With the retraction, the objective function f defined in a neighborhood of $\mathbf{P}_n \in \mathcal{M}$ can be transformed into a function defined on the linear tangent space $T_{\mathbf{P}_n}\mathcal{M}$ and can be expressed as [17] $h\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right) = f\left(R_{\mathbf{P}_n}^{\mathcal{M}}\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right)\right)$ with $h(\mathbf{0}) = f(\mathbf{P}_n)$.

Instead of directly optimizing $\mathbf{P}_n$ on the nonlinear manifold $\mathcal{M}$, $h\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right)$ paves the way for searching for a good point on the linear tangent space, which facilitates the optimization on the nonlinear Riemannian manifold $\mathcal{M}$. The direction of an efficient $\boldsymbol{\xi}_{\mathbf{P}_n}$ is usually related to the negative Riemannian gradient. Let $\check{\mathbf{R}}_{kn} = \hat{\mathbf{H}}_{kn}\sum_{\ell\in\mathcal{U}}\mathbf{P}_{\ell n}\mathbf{P}_{\ell n}^H\hat{\mathbf{H}}_{kn}^H + \sigma_z^2\mathbf{I}_{M_r}$. Then, the Euclidean gradient of the $f(\mathbf{P}_n)$ is given by $\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_n) = (\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_{1n}), \cdots, \mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_{Un}))$, where $\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_{kn}) = -2\sum_{\ell\in\mathcal{U}}\hat{\mathbf{H}}_{\ell n}^H\check{\mathbf{R}}_{\ell n}^{-1}\hat{\mathbf{H}}_{\ell n}\mathbf{P}_{kn} + 2\sum_{\ell\neq k}\hat{\mathbf{H}}_{\ell n}^H\mathbf{R}_{\ell n}^{-1}\hat{\mathbf{H}}_{\ell n}\mathbf{P}_{kn}$. The Riemannian gradient of $f(\mathbf{P}_n)$ on $T_{\mathbf{P}_n}\mathcal{M}$ is given by

$$\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n) = \mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_n) - \lambda\mathbf{P}_n, \tag{17}$$

where $\lambda = \frac{1}{P}\Re\left\{\mathrm{tr}\left(\mathbf{P}_n^H \mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_n)\right)\right\}$.

To accelerate the convergence speed, it is suggested that the previous update direction should be taken into consideration [17]. Let $\boldsymbol{\xi}_{\mathbf{P}_n} \in T_{\mathbf{P}_n}\mathcal{M}$ and $\boldsymbol{\zeta}_{R_{\mathbf{P}_n}^{\mathcal{M}}(\boldsymbol{\eta}_{\mathbf{P}_n})} \in T_{R_{\mathbf{P}_n}^{\mathcal{M}}(\boldsymbol{\eta}_{\mathbf{P}_n})}\mathcal{M}$ denote two tangent vectors in different tangent spaces. The vector transport denoted by $\mathcal{T}_{\boldsymbol{\eta}_{\mathbf{P}_n}}^{\mathcal{M}}\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right)$ can transport the tangent vector $\boldsymbol{\xi}_{\mathbf{P}_n}$ from $T_{\mathbf{P}_n}\mathcal{M}$ to the tangent space $T_{R_{\mathbf{P}_n}^{\mathcal{M}}(\boldsymbol{\eta}_{\mathbf{P}_n})}\mathcal{M}$, facilitating the operations between the tangent vectors in different tangent spaces. For the Riemannian submanifold $\mathcal{M}$, $\mathcal{T}_{\boldsymbol{\eta}_{\mathbf{P}_n}}^{\mathcal{M}}\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right)$ can be achieved by orthogonally projecting $\boldsymbol{\xi}_{\mathbf{P}_n}$ onto $T_{R_{\mathbf{P}_n}^{\mathcal{M}}(\boldsymbol{\eta}_{\mathbf{P}_n})}\mathcal{M}$. Let $\mathbf{P}_n^{\mathrm{new}} = R_{\mathbf{P}_n}^{\mathcal{M}}\left(\boldsymbol{\eta}_{\mathbf{P}_n}\right)$, the vector transport takes the form of $\mathcal{T}_{\boldsymbol{\eta}_{\mathbf{P}_n}}^{\mathcal{M}}\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right) = \boldsymbol{\Pi}_{T_{\mathbf{P}_n^{\mathrm{new}}}\mathcal{M}}^{T_{\mathbf{P}_n}\mathcal{M}}\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right) = \boldsymbol{\xi}_{\mathbf{P}_n} - \varphi\mathbf{P}_n^{\mathrm{new}}$, where $\varphi = \frac{1}{P}\Re\{\mathrm{tr}((\mathbf{P}_n^{\mathrm{new}})^H\boldsymbol{\xi}_{\mathbf{P}_n})\}$, and $\boldsymbol{\Pi}_{T_{\mathbf{P}}\mathcal{M}}^{T_{\mathbf{P}}\mathcal{N}}(\boldsymbol{\xi}_{\mathbf{P}})$ denotes the orthogonal projection of the tangent vector $\boldsymbol{\xi}_{\mathbf{P}}$ from $T_{\mathbf{P}}\mathcal{N}$ to $T_{\mathbf{P}}\mathcal{M}$.

IV. Riemannian Conjugate Gradient Method for Precoder Design

A. Riemannian Conjugate Gradient Method

For the optimization on the nonlinear $\mathcal{M}$, the retraction (16) is utilized to transform the function f defined on $\mathcal{M}$ into the function $\mathrm{h}\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right)$ defined on the linear tangent space, where $\boldsymbol{\xi}_{\mathbf{P}_n} = \alpha\boldsymbol{\eta}_n$ with $\boldsymbol{\eta}_n$ and α being the search direction and step length, respectively. For notational clarity, any notation with the superscript p denotes the notation in the p-th outer iteration, and any notation with the superscript pair (p,q) represents the notation in the q-th iteration of searching for the step length during the p-th outer iteration. The step length α^p can be effectively achieved via the backtracking method [17]. The update formula when searching for the step length is

$$\mathbf{P}_n^{p,q} = \rho^{p,q}\left(\mathbf{P}_n^p + \alpha^{p,q}\boldsymbol{\eta}_n^p\right), \tag{18}$$

where $\alpha^{p,q}$ is the trial step length in the (p,q)-th iteration. (18) is repeated until a satisfactory $\alpha^p = \alpha^{p,q}$ is obtained with $\mathbf{P}_n^{p+1} = \mathbf{P}_n^{p,q+1}$ being the input of the next iteration. During this process, the objective function needs to be compared repeatedly until a sufficient decrease is obtained. The objective function in the (p,q)-th iteration can be rewritten as

$$f(\mathbf{P}_n^{p,q}) = -\sum_{k\in\mathcal{U}} w_k \mathcal{R}_{kn}^{p,q}, \tag{19}$$

where $\mathcal{R}_{kn}^{p,q} = \mathrm{logdet}\left(\check{\mathbf{R}}_{kn}^{p,q}\right) - \mathrm{logdet}\left(\mathbf{R}_{kn}^{p,q}\right)$.

The Riemannian conjugate gradient and can be expressed as

$$\boldsymbol{\eta}_n^p = -\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p) + \beta^{p-1}\mathcal{T}_{\alpha^{p-1}\boldsymbol{\eta}_n^{p-1}}\left(\boldsymbol{\eta}_n^{p-1}\right). \tag{20}$$

β^p is chosen as the modified Polak and Ribière parameter (PRP) to avoid jamming [18] and is given by $\beta^p = \max(0, \min(\beta_{\mathrm{PRP}}^p, \beta_{\mathrm{FR}}^p))$, where $\beta_{\mathrm{FR}}^p = \frac{g_{\mathbf{P}_n^p}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p), \mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p)\right)}{g_{\mathbf{P}_n^{p-1}}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^{p-1}), \mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^{p-1})\right)}$. Let $\boldsymbol{\nu}_n^p = \mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p) - \mathcal{T}_{\alpha^{p-1}\boldsymbol{\eta}_n^{p-1}}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f\left(\mathbf{P}^{p-1}\right)\right)$, β_{PRP}^p is given by $\beta_{\mathrm{PRP}}^p = \frac{g_{\mathbf{P}^p}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p), \boldsymbol{\nu}_n^p\right)}{g_{\mathbf{P}_n^{p-1}}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f\left(\mathbf{P}_n^{p-1}\right), \mathrm{grad}_{\mathcal{M}} f\left(\mathbf{P}_n^{p-1}\right)\right)}$.

B. Efficient Algorithm Implementation

With the search direction (20), the problem (15) can be directly solved by the conventional RCG method [19]. Nevertheless, the objective function (19) needs to be calculated repeatedly during the inner iteration for comparing the value of the objective function (19), implying that the computational complexity is dependent on the number of inner iterations, which is still unacceptable in practice. In the following, we will show that the computational complexity can be significantly decreased and is irrelevant to the number of inner iterations by precomputing some elements. Let $\hat{\mathbf{E}}_{k\ell}^{p,q} = \hat{\mathbf{H}}_{kn}\mathbf{P}_{\ell n}^{p,q} \in \mathbb{C}^{M_r\times d_k}$. With (18), $\hat{\mathbf{E}}_{k\ell}^{p,q}$ can be computed by

$$\hat{\mathbf{E}}_{k\ell}^{p,q} = \rho^{p,q}\hat{\mathbf{H}}_{kn}\left(\mathbf{P}_{\ell n}^p + \alpha^{p,q}\boldsymbol{\eta}_{\ell n}^p\right), \tag{21}$$

indicating that $\hat{\mathbf{E}}_{k\ell}^{p,q}$ can be effectively computed during the inner iteration with the precomputed $\hat{\mathbf{E}}_{k\ell}^p = \hat{\mathbf{H}}_{kn}\mathbf{P}_{\ell n}^p$ and $\bar{\mathbf{E}}_{k\ell}^p = \hat{\mathbf{H}}_{kn}\boldsymbol{\eta}_{\ell n}^p$, which are all irrelevant to the index of inner iterations. With (21), $\mathbf{R}_{kn}^{p,q}$ and $\check{\mathbf{R}}_{kn}^{p,q}$ can be rewritten as

$$\mathbf{R}_{kn}^{p,q} = \sum_{\ell\neq k}\hat{\mathbf{E}}_{k\ell}^{p,q}\left(\hat{\mathbf{E}}_{k\ell}^{p,q}\right)^H + \sum_{\ell\neq k}\boldsymbol{\Phi}_{kn}\left(\mathbf{P}_{\ell n}^{p,q}\left(\mathbf{P}_{\ell n}^{p,q}\right)^H\right) + \sigma_z^2\mathbf{I}_{M_r} \tag{22}$$

and

$$\check{\mathbf{R}}_{kn}^{p,q} = \sum_{\ell\in\mathcal{U}}\hat{\mathbf{E}}_{k\ell}^{p,q}\left(\hat{\mathbf{E}}_{k\ell}^{p,q}\right)^H + \sum_{\ell\in\mathcal{U}}\boldsymbol{\Phi}_{kn}\left(\mathbf{P}_{\ell n}^{p,q}\left(\mathbf{P}_{\ell n}^{p,q}\right)^H\right) + \sigma_z^2\mathbf{I}_{M_r}, \tag{23}$$

respectively. With the precomputed $\hat{\mathbf{F}}_{k\ell}^p$, $\bar{\mathbf{F}}_{k\ell}^p$, $\tilde{\mathbf{F}}_{k\ell}^p$, $\hat{\mathbf{E}}_{k\ell}^p$ and $\bar{\mathbf{E}}_{k\ell}^p$, $\mathbf{R}_{kn}^{p,q}$ (22) and $\check{\mathbf{R}}_{kn}^{p,q}$ (23) can be derived through a weighted sum

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of known matrices, showing that the objective function can be efficiently computed during the inner iteration.

After reaching a good point in the (p,q)-th iteration, $f\left(\mathbf{P}_n^{p+1}\right) = f\left(\mathbf{P}_n^{p,q}\right)$, $\hat{\mathbf{E}}_{k\ell}^{p+1} = \hat{\mathbf{E}}_{k\ell}^{p,q}$, $\hat{\mathbf{F}}_{k\ell}^{p+1} = \mathbf{\Phi}_{kn}\left(\mathbf{P}_{\ell n}^{p,q}\left(\left(\mathbf{P}_{\ell n}^{p,q}\right)^H\right)\right)$, $\mathbf{R}_{kn}^{p+1} = \mathbf{R}_{kn}^{p,q}$ and $\check{\mathbf{R}}_{kn}^{p+1} = \check{\mathbf{R}}_{kn}^{p,q}$ can be seamlessly utilized as the input of the next iteration. Then, the Euclidean gradient in the $(p+1)$-th iteration can be further written as

$$\begin{aligned}\mathrm{grad}_{\mathcal{N}} f\left(\mathbf{P}_{kn}^{p+1}\right) = &-2\sum_{\ell\in\mathcal{U}}\left(\hat{\mathbf{H}}_{\ell n}^H\left(\check{\mathbf{R}}_{\ell n}^{p+1}\right)^{-1}\hat{\mathbf{E}}_{\ell k}^{p+1}\right)\\ &+2\sum_{\ell\neq k}\left(\hat{\mathbf{H}}_{\ell n}^H\left(\mathbf{R}_{\ell n}^{p+1}\right)^{-1}\hat{\mathbf{E}}_{\ell k}^{p+1}\right),\end{aligned} \tag{24}$$

where $\hat{\mathbf{E}}_{\ell k}^{p+1}$, $\check{\mathbf{R}}_{\ell n}^{p+1}$ and $\mathbf{R}_{\ell n}^{p+1}$ all can be retrieved from the outputs of the previous iteration without incurring additional complexity.

The RCG method for precoder design is provided in Algorithm 1, where r and c are typically chosen as 0.5 and 10^{-4}, respectively.

Algorithm 1 RCG method for precoder design.

Require: Riemannian submanifold $\mathcal{M}$; Riemannian metric $g_{\mathbf{P}}^{\mathcal{M}}(\cdot)$; Real-valued function f; Retraction $R_{\mathbf{P}}^{\mathcal{M}}(\cdot)$; Vector transport $\mathcal{T}_{\boldsymbol{\eta}_n}^{\mathcal{M}}(\cdot)$; initial step length $\alpha^0 > 0$; $r \in (0,1)$; $c \in (0,1)$

Input: $\hat{\mathbf{H}}_{kn}$, $\tilde{\mathbf{\Omega}}_{kn}, \forall k \in \mathcal{U}$; initial point $\mathbf{P}^0$;

repeat
  Get $\hat{\mathbf{E}}_{k\ell}^p = \hat{\mathbf{H}}_{kn}\mathbf{P}_{\ell n}^p$ for $k,\ell \in \mathcal{U}$.
  Compute Euclidean gradient $\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_{kn}^p)$ with (24).
  Get Riemannian gradient $\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_{kn}^p)$ with (17).
  Update the search direction $\boldsymbol{\eta}_n^p$ with (20) and compute $\bar{\mathbf{E}}_{k\ell}^p = \hat{\mathbf{H}}_{kn}\boldsymbol{\eta}_{\ell n}^p, k,\ell \in \mathcal{U}$.
  while $f\left(\mathbf{P}_n^{p,q-1}\right) - f\left(\mathbf{P}_n^p\right) \geq c \times g_{\mathbf{P}^n}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f\left(\mathbf{P}_n^p\right), \alpha^{n,m-1}\boldsymbol{\eta}_n^p\right)$ **do**
    Set $\alpha^{p,q} \leftarrow r\alpha^{p,q-1}$ with $\alpha^{p,0} = \alpha^0$.
    Get $\mathbf{P}_n^{p,q+1}$ with (18) and $\hat{\mathbf{E}}_{k\ell}^{p,q}, i,j \in \mathcal{S}_U$ with (21).
    Get $f\left(\mathbf{P}_n^{p,q}\right)$ with (19), (22) and (23), $q \leftarrow q+1$.
  end while
  Set $\mathbf{P}_n^{p+1} \leftarrow \mathbf{P}_n^{p,q}$, $\hat{\mathbf{E}}_{k\ell}^{p+1} \leftarrow \hat{\mathbf{E}}_{k\ell}^{p,q}$, $p \leftarrow p+1$.
until convergence

C. Computational Complexity

Let N_{out}, $N_S = \sum_{k\in\mathcal{U}} d_k$, and $N_R = UN_r$ denote the total numbers of the outer iterations, data streams and user antennas, respectively. Then, we focus on the complexity analysis of each iteration. During each iteration of the RCG method, the Euclidean gradient $\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}^p)$ needs to be calculated once according to (24), where $\hat{\mathbf{E}}_{\ell k}^p$ can be directly obtained from the $(p-1)$-th iteration. The computational complexity of $\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}^p)$ is $O(M_t N_S N_R + M_t^2 N_S)$. Before entering the inner iteration for searching for the step length, $\bar{\mathbf{E}}_{k\ell}^p$ is precomputed to facilitate the computation of the objective function during the inner iteration. During the inner iteration, the objective function needs to be computed per inner iteration, whose input $\mathbf{R}_{kn}^{p,q}$ and $\check{\mathbf{R}}_{kn}^{p,q}$ can be directly obtained with (22), (23). In addition, the complexity of computing $\log\det(\cdot)$ in the objective function can be omitted as the dimension of the input is M_r. Therefore, the computational complexity of the RCG method is $O(2M_t N_S N_R)$. The method proposed in [20] aims at maximizing the upper bound of the ergodic WSR based on the channel estimated from the pilot segments via minorize-maximization (MM) methodology, whose computational complexity is $O(M_t^2\log(N_t) + 2N_t\log(M_t)N_S + M_t^2 N_S + M_t^3/2)N_{\mathrm{out}}$. Compared with this MM based method, our proposed RCG method avoids the inverse of large dimensional matrices and exhibits a lower complexity.

V. Simulation Results

To establish a simulation environment that reproduces real-world characteristics of the HSR, we adopt the prevalent QuaDRiGa channel model [21] to generate a simulation scenario, where the "3GPP 37.885 Highway LOS" is considered to simulate the viaduct scenario. The HSR is programmed to travel eastward along the x-axis with $U = 20$ UTs uniformly spaced at 10-meter intervals on the HSR. The initial position of the first UT is set at coordinates (-200,0) and the vertical BS-track distance is set to 200 m with the BS positioned at (0,-200). The time duration of each subframe is set to 1 ms and the subcarrier spacing is set to 60 KHz. More detailed simulation parameters are available in Table I.

First of all, we compare the normalized mean-squared error (NMSE) performance of the predicted channel (10) and the pilot segment channel (9) in Fig. 2, where the sixth OFDM index is the pilot symbol. By observing Fig. 2, we can see that directly applying the pilot segment channel to the data symbol has a much poorer NMSE performance compared with the predicted channel. In addition, the NMSE performance of the predicted channel improves as the fine factor F_{do} increases.

To show the superiority of the proposed RCG method for precoder design with predicted channel, we compare the WSR performance of the RCG method with those of other approaches at the fourteenth OFDM symbol after the pilot symbol, which is 0.25 ms away from the pilot symbol. For clarity, the precoders designed by the RCG method with the pilot segment channel (9) and the predicted channel (11) are termed as the RCG-pilot precoder and the RCG-predicted precoder, respectively. As shown in Fig. 3, the RCG-predicted precoder with $F_{\mathrm{do}} = 2$

TABLE I
Detailed Simulation Parameters

f_c	4.8 GHz	v	300 km/h
M_t	128	M_r	2
N_c	2048	N_g	144
N_v	120	Δf	60 KHz
N_b	4	N_s	14
$P_u, \forall u \in \mathcal{U}$	7	Ricean K-factor	7 dB
Δ_r	$\frac{1}{2}$	Δ_t	$\frac{1}{2}$
F_r	1	F_t	2
F_{de}	2	F_{do}	2
$d_k, \forall k \in \mathcal{U}$	2	$w_k, \forall k \in \mathcal{U}$	1
Height of each BS	25 m	Height of each UT	1.5 m

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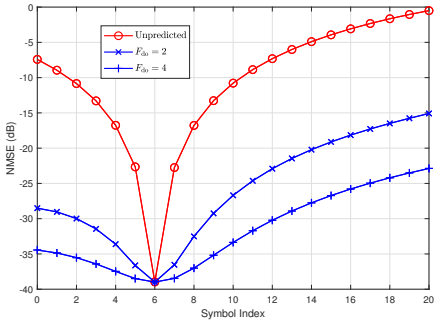


Fig. 2. The comparison of NMSE performance versus OFDM symbol index.

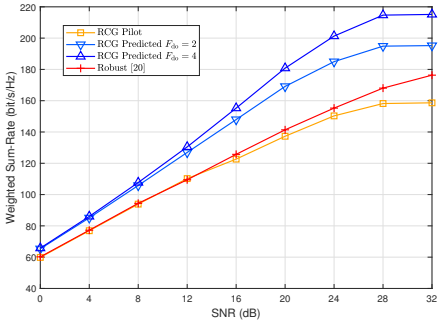


Fig. 3. The WSR performance of the RCG method at the fourteenth predicted OFDM symbol.

has a much better performance compared with that of the RCG-pilot precoder, showing the advantage of our predicted channel. In addition, the RCG-predicted precoder exhibits significant performance gain compared with that of the robust precoder designed by the method proposed in [20], further validating the superiority of the proposed channel model and the RCG method. We also observe that the WSR performance of the RCG-predicted precoder with $F_{\rm do}=4$ outperforms that of the RCG-predicted precoder with $F_{\rm do}=2$, primarily due to the enhanced accuracy of CSI resulting from finer sampling.

VI. Conclusion

In this paper, we investigated the matrix manifold precoder design for massive MIMO HSR communications with channel prediction. The QB based channel model was established for acquiring a more accurate CSI. With the matrix manifold optimization, the ergodic WSR-maximization problem was formulated on the Riemannian submanifold formed by the precoders satisfying TPC, and the Riemannian ingredients were derived. With the proposed QB model and the derived Riemannian ingredients, the RCG method was proposed for solving the unconstrained optimization problem on the Riemannian submanifold. The complexity analyses demonstrated the computational efficiency of the proposed RCG method. The simulation results verified the high accuracy of the predicted channel and the superiority of the RCG method for precoder design for the massive MIMO HSR system.

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